

Towards Agricultural Cyberphysical AI: A New Multimodal Hyperspectral Remote Sensing Platform

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Abstract—Hyperspectral Imaging (HSI) is becoming an increasingly popular technology, especially in the domain of precision agriculture, due to its ability to detect plant health problems early on. Hyperspectral cameras are typically expensive, closed systems designed purely for raw data collection. This paper presents an open-source, bespoke handheld/UAV/UGV-mountable hyperspectral imaging payload for on-board data processing and modelling, as a step towards a full Cyber Physical System (CPS) for agriculture. It utilises Ximea XiSpec snapshot cameras in visible and near-infrared (MQ022HG-IM-SM4X4-VIS3, MQ022HG-IM-SM5X5-NIR2) as the hyperspectral sensors and a Jetson Orin Nano computer hosting a software stack based on the Robot Operating System 2 (ROS2) Humble distribution. The ROS2 design allows easy integration into existing robotics and CPS utilising the same architecture. This paper contributes to existing literature by providing the described hardware/software system open source, allowing researchers to replicate and add their own algorithms for on-board data processing. The system was trialled in the field with power consumption of 10.99 ± 0.08 W whilst collecting 2xRGB, 2xHSI(raw), 1xDepth, IMU, GPS and spectrometer data at 20 FPS. It is useful for anyone with access to these cameras wishing to develop their own payload, for anyone wanting to purchase a new hyperspectral imaging system wanting full control over the system for custom pipelines, or for anyone developing their own hyperspectral sensor and wanting to start from an open-source, pre-tested system architecture.

Index Terms—Hyperspectral, UAV, Platform, Agriculture, Cyber Physical System

I. INTRODUCTION

The use of Hyperspectral Imaging (HSI) in agriculture is a well-researched, popular approach [1] resulting in highly effective methods for High Throughput Phenotyping (HTP), Chlorophyll content analysis, fungal disease detection, drought stress detection, weeds detection and management and crop classification, without requiring any invasive sensors and able to quickly collect data over wide areas. The reflectance data is frequently converted into lower dimension indices like the Normalised Difference Vegetation Index (NDVI [2]) to serve

as features into plant models. The complex nature of the radiometric calibrations, together with the proprietary nature of most commercial HSI systems, has lead to a standard procedure of collecting data and processing data as two separate stages. This results in a potentially significant loss of time before any predictions based on the processed data can be acted on, such as targeted application of pesticides to infested plants. Visual-based inspection methods also suffer on early detection periods compared to more invasive methods that can detect diseases in their incubation periods [3], therefore reducing processing time is a key goal for increasing time for corrective actions.

We propose a solution to this by designing an open source hyperspectral image and data processing platform that can be mounted on Unmanned Air Vehicles (UAVs), Unmanned Ground Vehicles (UGV) and handheld gimbals, as shown in Fig. 2. The motivations behind open-sourcing the project is to help enable others to build similar systems, learning and growing together, with the ultimate aim to create functional autonomous agricultural Cyber Physical Systems (CPS) as shown in Fig. 1. Ximea XiSpec snapshot visible and near-infrared cameras [4] (MQ022HG-IM-SM4X4-VIS3, MQ022HG-IM-SM5X5-NIR2) are used as the hyperspectral imaging sensors which provide a good balance between weight, complexity, spatial and spectral resolution. The software is hosted on a Jetson Orin Nano computer running Jetpack 6.2.1 (Ubuntu 22.04 based) and built within a Robot Operating System 2 (ROS2) Humble architecture. The ROS2 design allows easy integration into existing robotics and CPS utilising the same architecture. The system provides an open source platform for programmers to add their own algorithms for on-board data processing by adding ROS2 nodes, that can also be developed in a distributed manner, enabling complex CPS to be developed. This reduces design and testing time for researchers, as well as enabling further progress towards on-

TABLE I

DRONE-MOUNTED HYPERSPECTRAL IMAGING SYSTEMS: PS=PUSH-BROOM, SS=SNAPSHOT, LF=LIGHT FIELD, SBC=SINGLE BOARD COMPUTER. COSTS ARE APPROXIMATED FROM QUOTES AND ONLINE MATERIAL, LOW < £33K, MID = £33K-£66K, HIGH = £67K-£100K+ (VAT INCLUSIVE). * LINE FRAME RATE HAS BEEN DIVIDED BY 270 TO GIVE A ROUGH EQUIVALENT FOR A TYPICAL SNAPSHOT FPS. ** HIGH RESOLUTION ACHIEVABLE THROUGH "AI-BASED SPECTRAL ESTIMATION TECHNIQUE".

Company	Model Name	Imaging Type	On-board Processing	Open Source	Spectral Range [nm]	Resolution [px]	Frame Rate [Hz]	Bands	Weight [g]	Cost
Specim	AFX10(+SBC)	PS	No	No	400 – 1000	1024	0.55*	224	2100	Mid
Headwall	Nano-Hyperspec	PS	No	No	400 – 1000	640	1.1*	270	1290	High
Resonon	Pika L	PS	No	No	400 – 1000	900	0.92*	281	1830	Mid
BaySpec	OCI-F-120(SBC)	PS	Yes	No	400 – 1000	800	0.22*	120	650	Low
BaySpec	OCI-D2000(+SBC)	SS	Yes	No	475 – 875	500 × 270	120	40	450	Mid
Cubert	ULTRIS X20	SS + LF	No	No	350 – 1000	410 × 410	4	164	1200	Mid
Spectricon	MUSE9-DR	SS	Yes	No	400 – 1000	4096 × 2160**	24	22	450	Mid
Ximea Xispec (Ours)	VIS3, NIR2(+SBC)	SS	Yes	Yes	400 – 1000	512 × 272	20	41	840	Low

board plant disease detection and prevention.

Similar DIY approaches are documented in prior literature ranging from cheap, fully custom imaging sensors [5] that come with their own software but which do not try to provide high-throughput or on-board processing. Others provide more complex, holistic systems [6] but do not open source the design. Some are created to solve specific issues like smile aberration [7] but are not appropriate for field work. We believe our approach of combining off-the-shelf, reliable HSI sensors with customisable, open-source software provides a reasonably affordable and fully flexible platform that will appeal to researchers.

Currently available commercial HSI systems are summarised in Table I. This shows that this system is not only one of the cheapest, it is also one of only a few that allow on-board processing and the only one that provides a fully customisable open-source platform.

The key contributions of this research are:

- 1) Open-sourced ROS2 Humble software stack for multi-modal hyperspectral imaging and processing.
- 2) 3D Design Files.
- 3) Materials and costing breakdowns.
- 4) System demonstration results from showing capabilities of the multimodal system.

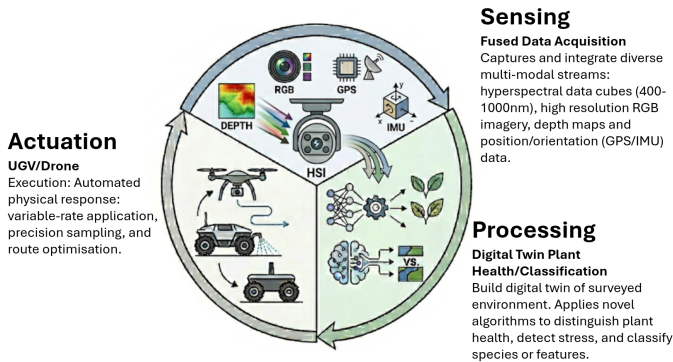


Fig. 1. A basic layout for an agricultural CPS. The platform described in this research is designed to provide both the sensing and processing elements, and built to easily communicate with the final actuation element by being built upon ROS2.

II. BACKGROUND

Hyperspectral sensors enable small, discrete wavelength ranges of light to be captured independently. In contrast to the 3 channels of RGB images, hyperspectral images can consist of hundreds of channels, known as spectral bands, spread over a wider wavelength range, typically from visible to infrared. In the case of hyperspectral these bands are narrow and contiguous whereas in multispectral images these bands can be much wider, fewer in number, and can be non-contiguous. The results are a spectral curve per pixel of an image that, when properly processed, reveals intrinsic reflective properties of the material being imaged, referred to as a spectral signature. This technology emerged from spectroscopy as a practical imaging technology in the late 1970's where push-broom methods were first available [8]. More recent advancement in micro-scale manufacturing has allowed individual pixels be coated in bandpass materials, like Fabry-Perot filters, that allow full "snapshot" style hyperspectral images (HSI) that are more similar to standard RGB sensors [9]. Though typically unable to match the spectral resolution of push-broom methods due to having fewer bands, the snapshot approach allows instant spatial information without post-processing.

Analysis of plant matter is highly linked to the chlorophyll concentrations in the subject [10]. Chlorophyll is highly absorbent in the visible region (400-750 nm) but has little effect on the NIR region (750-1400 nm) in which structural features of the target dominate. For chlorophyll-containing subjects this results in a characteristic "red edge", beginning at around 700 nm, where the spectral curve reflectance increases significantly. The red edge phenomena has led to several spectral indices designed specifically to quantify this effect and therefore predict plant matter density as well as health, one of the most common being Normalised Vegetation Difference Index [11] (NDVI).

Using hyperspectral sensors from satellites allows larger areas to be covered with a lower cost per area, and more consistent tracking over time. Whereas UAV/ground inspections come at a higher cost per area they outperform satellite system in spatial details, and can therefore be used to detect potential issues earlier such as disease and pests [12] [13]. As well as being able to be used on cloudy days, in indoor environments, and at shorter intervals of time.

The main advantage of hyperspectral imaging is to recover the inherent reflectance of the material being imaged. This reflectance is recovered through complex radiometric processing [14] [15] and represents an intrinsic property of the material which can therefore be used to compare the same types of material across different illumination environments. This is only possible by undergoing a thorough calibration procedure based on intrinsic image decomposition (IID). IID is a problem defined whereby the image is equivalent to the product of the illumination and the reflectance of the objects in the images, as in Eq. (1):

$$I(i, j, \lambda) = R(i, j, \lambda)L(i, j, \lambda) \quad (1)$$

where $I(i, j, \lambda)$ denotes the measured spectral intensity at pixel coordinates (i, j) and wavelength λ , $R(i, j, \lambda)$ denotes the spectral reflectance of the observed surface, and $L(i, j, \lambda)$ denotes the spectral illumination incident upon the surface. This is an ideal form of the problem which does not take into account the realities of imperfect capturing technology, for which there is a thorough calibration procedure. A more detailed discussion of the calibration procedures is available in [16] [14] [15].

The requirements of the calibration stage mean that the system requires user actions to place the system in a dark state (lens covers on) and set up the white reference for the white reference image. In addition, as the dark current is dependent on exposure/integration time, any change in this value must require a suitable dark image, or dark images which cover the upper and lower integration times and can be used to interpolate values. Depending on the use case the numerical aperture and focus may also need to be hand adjusted. For these reasons the system being designed must be able to provide live image feedback as well as control communication from a remote user interface.

ROS2 [17] is primarily a middleware architecture for developing distributed robotic systems, providing various development and debugging tools, as well as standardised and robust messaging protocols to allow various types of command input as well as feedback. It is an open source system providing multiple distributions that have long time service support. Its features enable rapid development of complex robotic systems while maintaining modularity and scalability which has made it popular in both research and industry [18]. In this work, ROS2 is used as the underlying middleware to coordinate sensor integration, perception, and control components within a modular architecture and allow easy integration to existing ROS2 systems.

III. METHODOLOGY

A. Hardware

The system is designed to be lightweight and fit within the footprint of a standard consumer camera, be constructable from off-the-shelf parts with minimal effort, and run on both desktop and battery powered setups. A complete bill of materials is shown in Table II. The system has multiple M3



Fig. 2. Top Left: Payload mounted on Gremsy T3 Gimbal. Top Right/Bottom Left: Gimbal + DJI Matrice M600 Drone. Bottom Right: DJI RS 3 Mini handheld gimbal during a white referencing procedure in the field.

holes to allow fitting of dovetail slide adaptors for connecting to different systems. The case is 3D printed in black PLA filament, fitting in a total footprint of 90 x 100 x 150 mm, weighing a total of 842 g with all peripherals installed, which could be further reduced by reducing infill of the 3D printed case from 100%. Due to the necessity for a stable hyperspectral sensor temperature and the excess heat created by both the cameras and the Jetson a dual fan active cooling system is powered from the Jetson's GPIO pins 2 & 6. In addition to the hyperspectral sensors (MQ022HG-IM-SM4X4-VIS3 & MQ022HG-IM-SM5X5-NIR2) the system accommodates a stereo RGB sensor with on-board IMU (Waveshare IMX219-83) connected to the 2x CSI connectors and GPIO pins 3 & 5, as well as a low-cost downwelling sensor (MIKROE-4165) connected to GPIO pins 1, 9, 27 & 28. The Jetpack OS runs off a 64Gb SD card, with a 1TB SSD for data storage. 3D printable parts are available through the Github for this research [19]. Fig. 2 shows the system in its various use modes and Fig. 3 shows an exploded view of the mechanical design.

The system is powered from the Jetson which is in turned powered from a 5.5 mm x 2.5 mm jack. This can be powered from the supplied desktop power supply included in the developers kit, or a Power Delivery (PD)-enabled battery charger with PD cable (Jetson Orin Nano does not have native PD communication functionality). A 2-pin JST female connector has been soldered onto the terminals of the jack connector on the Jetson board to enable power via the Gremsy gimbal power out connector from the drone. For quick access the case uses common flip latches, but for extra security whilst in the air there are four M3x8 SHC holes to securely lock the case. These are designed to be used with soldered-in M3 brass thread inserts but can be used with self-tapping screws as well.



Fig. 3. Exploded view of the hardware design.

B. Software Framework and Data Pipeline

The system is built in a ROS2 workspace consisting of multiple subsystems: controller, sensors, processing, and monitoring. Each contain multiple C++/python node packages with limited function scope, in order to maintain scalability and prevent any blocking code. The full software stack is available on Github [19] and a system overview is shown in Fig. 4.

Sensors: Contains driver nodes for hyperspectral camera frame-grabbing, stereo camera frame-grabbing, IMU, GPS and spectrometer raw data recording. The `hsi_camera_driver` is a single node that can be launched with different parameters allowing it to record data from the slightly different VIS and NIR sensors. It uses a combination of the Ximea xiAPI [20], and IMEC HSI_API for framegrabbing, sensor temperature measurements, and saving the raw image/metadata files in the predefined IMEC method. This node also handles any camera interactions such as integration time, auto exposure, saving

TABLE II
CONSOLIDATED BILL OF MATERIALS REQUIRED TO BUILD THE DESCRIBED SYSTEM. THE MOST COSTLY ELEMENTS ARE THE TWO HYPERSPECTRAL SENSORS, IN COMPARISON BUILDING THE SYSTEM AROUND THEM IS FAIRLY CHEAP AND ALLOWS A LOT OF FLEXIBILITY.

Part Name	Part Number	Supplier	Price (£/ea)
HSI Camera + Kit Visible Range	MQ022HG-IM-SM4X4-VIS3	Ximea/Imec	14,500.00
HSI Camera + Kit Near-IR Range	MQ022HG-IM-SM5X5-NIR2	Ximea/Imec	14,500.00
Gremsy T3V3 to M600 CAN Cable	Gremsy-Cable DJI FC	3DXR	25.00
Gremsy T3V3 Gimbal	GREMSY-GIMBAL-T3V3	3DXR	1749.00
DJI RS 3 Mini Gimbal	DJI RS 3 Mini	DJI Store UK	219.00
Waveshare IMX219 Stereo Camera	B086JTDG25	Amazon	60.49
USB-C PD to DC Converter Cable	ASIN: B0GD716J8J	Amazon	7.99
65W 20KmAh PD Power Bank	B0FWR98WX5	Amazon	36.99
USB-C to USB-A Right Angle	B0BLMWH15B	Amazon	4.99
USB-A to USB-A Right Angle	B079CC2SGB	Amazon	4.99
Black PLA Filament (2Kg)	N/A	Amazon	17.02
25mm Focal Lens VIS NIR x2	67-715	Edmund Optics	372.00
Lexar M.2 1TB SSD	267-6408	RS	114.60
Nvidia Jetson Orin Nano	264-7384	RS	259.01
DFRobot USB GPS Receiver	247-3251	RS	11.53
AS7341 Spectral Colour Sensor	WAV-19564	PiHut	19.20
StarTech USB 3.0 A to Micro-B x2	186-3203	RS	20.88
RS PRO Toggle Switch (DPDT)	734-7044	RS	4.03
RS PRO Push Button Switch	734-6735	RS	2.69
Kingston 64GB MicroSD Card	186-3203	RS	8.84
Toggle Catch Lock	B0CLY2RL9J	Amazon	4.46
		Approx Total	£32,335.46

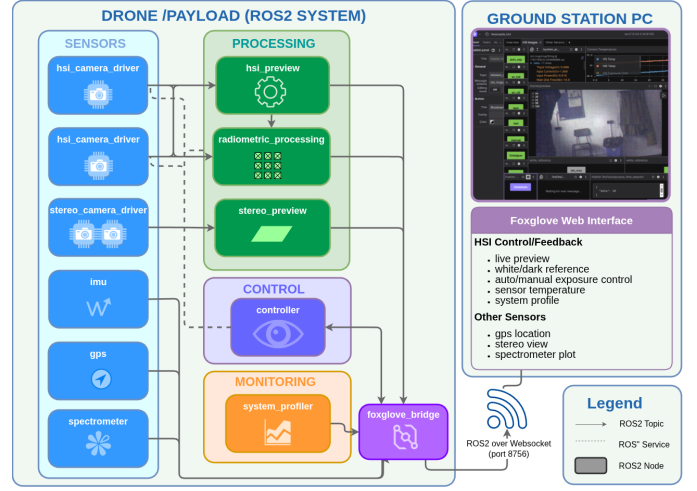


Fig. 4. Software system overview for the payload.

calibration images, and starting timelapses. Upon launching the camera nodes the context folder structure for the run is filled by duplicating the relevant template calibration and optical setup files. The cameras are then initialised in a continuous acquisition mode which overlaps the acquisition of the frame with the data readout of the previous frame, giving the highest frame rate. Typical exposure times outdoors range from 1-30 ms enabling rates from the maximum 340 Frames Per Second (FPS) and lower. For our purposes a 20 Hz background acquisition is the maximum required rate to allow for camera focussing and timelapsing, faster FPS is prevented via a timed sleep period. Timelapse data is triggered on a timer within the acquisition loop. All acquired data is published in real-time on a `/hsi/<vis/nir>/raw` ROS2 topic, this raw data shows the full sensor readout without any spectral demosaicing, by performing max binning in the downstream processing nodes this raw data can be used to adjust camera focus and auto set integration times to avoid overexposure. The node is also subscribed to ROS2 services which allow commands to be sent from the remote UI, processed on the payload, and returned with a success/fail response. The camera node is the most complex node due to the interactions required for the calibration process, the IMU, GPS and Spectrometer nodes require no interaction and simply acquire then publish their respective data at a sampling rate defined at launch.

Processing: This subsystem contains nodes that process the raw data passed on from the sensors either with the aim of compressing the raw data for publishing across the network to the remote UI, radiometric calibration or further on-board processing. The `hsi_binned_preview` node takes the raw HSI topic data and first bins the data according to the subpixel size of the snapshot camera, either 4x4 or 5x5 depending on the range, then further bins by a set factor of 4 to reduce network bandwidth issues. The type of binning, average or max can be selected, as well as displaying in a pseudo RGB colour. The resulting images are published on the `/hsi/<vis/nir>/preview` topic and displayed in the Foxglove UI. Any large latencies

cause blank frames to be sent instead, notifying the user of the lag, which in normal running conditions should not occur. Similar operations are provided by the `stereo_binned_preview` node which grabs both the left and right RGB frames then average bins at a set factor. It also computes and publishes disparity between the frames using the OpenCV [21] StereoBM method, though this requires pre-calibration of the system, and so far has yielded unsuitable results due to the lack of synchronisation of the IMX219 cameras. The radiometric-processing node is still under development but will eventually process the recorded raw hsi data according to the saved context information. This uses the `HSI_API` which has been timed at ~ 2 FPS whilst running on the Jetson currently, though in the future we would like to rewrite this in order to utilise the onboard GPU for real-time processing. We envisage that < 20 FPS should be suitable for a UAV aerial mission. Further nodes can easily be added using the `ros2 pkg create` command, which can then subscribe to any of the raw sensor or post-processed data to provide custom functionality to the payload.

Monitoring: The monitoring subsystem contains a `system_profiler` node which interacts with the Jetson system to save and publish live system information to the remote UI. It pulls data from the INA3221 power monitor, onboard thermistors and other system data, providing updates on power, disk space, memory, CPU/GPU usage and onboard temperatures.

Controller: This subsystem contains a single node designed to provide overall control of the system. This includes creating the data run directory, relaying service messages for camera commands between the remote UI and the system, controlling rosbag recording and safely shutting down the system. The rosbag records all raw data topics (excluding binned images that are purely for live visualisation), with the option to throttle rates to reduce disk space. The hyperspectral images are saved in the rosbag in their raw image formats, though these do not include the metadata required for post processing via the `IMEC_API`. For that the `hsi_camera_driver` saves the HSI frames at the specified rate direct to disk. What data is saved can be customised in the controller code.

Remote GUI: The complexities of the radiometric calibration process necessitates a remote GUI for ensuring that camera focus, exposure and composition are correct on the system. The interface should have low latency and sufficient resolution to allow focussing. Foxglove was used [22] as an out-the-box solution for this purpose due to its freemium pricing structure that allows access to all the relevant functionality at the free tier. The ease of this solution, was another reason for choosing to utilise the ROS2 environment for this design. Fig. 4 includes a screenshot of a section of the user interface showing previews of the live hyperspectral camera output, camera sensor temperature, profiling data, as well as buttons for controlling the system. The new user would simply have to download the pre-made layout file contained in the Github repo, connect to the same network as the Jetson, which can be a laptop hotspot, and then initiate the connection via either the browser or desktop version of foxglove.

Post Processing: A jupyter notebook script is provided for

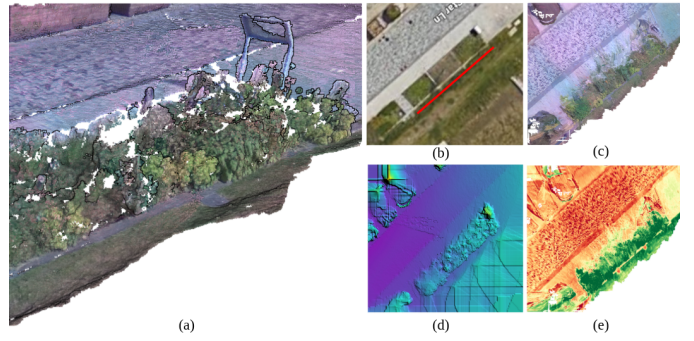


Fig. 5. Data generated from RGB images recorded on a test of the system at a flower patch outside the Newcastle University USB building: (a) point cloud (b) satellite image showing camera trajectory in red (c) orthomosaic (d) depth (e) NDVI

basic batch post processing. This unpacks the rosbag into a designated folder structure, radiometrically processes the HSI frames given the recorded calibration images and converts the left RGB images to a movie file for quick run previewing.

IV. SYSTEM TRIAL, RESULTS AND DISCUSSION

Once the system was ready to run it was tested in handheld mode on the Newcastle University Helix campus. All data was recorded with post processing performed on a desktop device. In the standard data collection mode of 20 FPS the system collects data at ~ 100 Mb/s with two (640, 480) RGB images, one (640, 480) depth image, two (2045, 1085) raw hyperspectral images, and the two (512, 272, 16/25) hyperspectral cubes at 1 FPS, as well as the other sensor data: GPS, IMU and spectrometer. The power usage was recorded at 10.99 ± 0.08 W during acquisition and with active cooling enabled, allowing for a theoretical operational time of ~ 9 hours on a 20,000 mAh battery pack. A number of simple runs were performed with initial white reference recorded using a Spectralon 95% reflective target in bright sunlight and low cloud cover. RGB images were processed in the open-source WebODM [23] software using Structure from Motion (SfM) techniques to produce the various images and point cloud shown in Fig. 5. In processing the RGB images we created an orthomosaic and point cloud from which we could extract the the depth and Normalised Difference Yellow Index (NDVI [24]) as an indicator of plant health without access to the NIR region data.

Results were radiometrically processed using the `HSI_API` to get the raw reflectance then further processed to produce the NDVI image shown in Fig. 6. The spectra of different targets in the image were extracted by taking the nearest 30 pixels and taking the average, using ± 1 *stdev* as the infill region. The diagram overall shows the unique benefits of HSI over RGB, and particularly near-infrared HSI. The RGB image for example shows clear colour differences between objects like the pavement and flower petal, but the plant objects are all quite a similar green. However, even in the visible HSI we can see that the grass, flower and petal spectra are quite similar but the two bush spectra are easily separable. The variance between

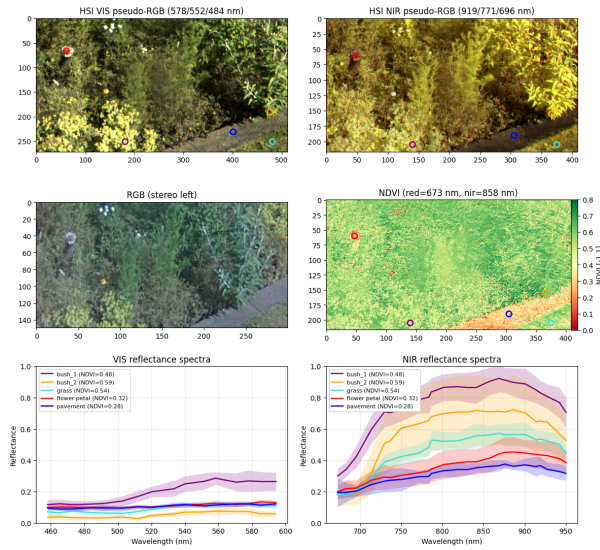


Fig. 6. The hyperspectral images (VIS and NIR) collected imaging a section of the same flower patch as show in Fig. 5. Radiometrically processed pseudocolour RGB images are shown in the top row, with a true RGB image from the left stereo camera, along with a processed NDVI image, and resulting average spectra from 5 points of 30 pixels in each of the VIS and NIR images, shown with infill representing ± 1 stdev.

the targets is even more pronounced in the near-infrared spectrum. This is before any actual dimension-reduction, feature extraction, spatial analysis or machine learning which would further enhance the ability to distinguish between the different targets [25]. The NDVI image is a dimension reduction method involving only two of the HSI bands that allows us to clearly separate plant matter from other objects. As an example use case the “bush_2” target from Fig 6 is actually Butterfly Bush (*Buddleja Davidii*) which is classified as a naturalised invasive non-native species, a segmentation algorithm based on the pixel spectra would be able to distinguish this from the other plants before evening taking into account spatial features. This showcases the usefulness of the hyperspectral approach that can be achieved with the hyperspectral payload we have built.

V. CONCLUSION AND FUTURE WORK

The system we are showcasing is an open-source hardware/software platform based on ROS2 Humble for the recording and live processing of multimodal data centred around the hyperspectral sensors. It is the only fully open mechanical/software design able to be built from off-the-shelf parts in the literature currently. In addition to being able to record visible and near-infrared snapshot hyperspectral images it also records depth, RGB, IMU, GPS and spectrometer data at desired frame rates tested up to 20Hz. This collection of data is particularly useful for SfM techniques as was shown in Section IV. From field trials the system has been shown to be effective in handheld mode for collecting data on local plants, whilst running on a battery pack, with a laptop connected via a local hotspot for controlling the system. This system [19] is of interest to other researchers as a jumping off point for hyperspectral research, specifically aimed at on-

board processing for applications in agriculture, such as cyber physical systems.

The next steps for this work involves improving upon the existing design in a few key areas: custom radiometric and on-board processing implementation for real-time data analysis, stereo camera synchronisation for improved depth perception, sensor fusion of all available data for improved environmental understanding of the system.

VI. ACKNOWLEDGEMENTS

Fig. 1 was created with the assistance of the artificial intelligence tool Gemini 3.1 Pro, which was used to support the visualisation and design process; the authors retain full responsibility for the final content and interpretation of the figure.

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